

AN2DL - Second Challenge Report

four_loops

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December 16, 2025

1 Abstract

This work emphasizes practical robustness and generalization rather than architectural novelty. This report presents a robust deep learning pipeline for histological image classification into four breast cancer subtypes: *HER2(+)*, *Luminal A*, *Luminal B*, and *Triple Negative*. The dataset is characterized by visual artifacts, limited labeled samples, and subtle inter-class differences. To address these challenges, we adopt a data-centric approach combined with large-scale architectural ensembling. Our final solution, the **Omni-Ensemble**, aggregates predictions from 23 models spanning EfficientNet, ResNet/ResNeXt, and DenseNet families. Through Smart Cropping, heavy augmentation, and 8-view Test-Time Augmentation (TTA), we achieve a public leaderboard score of **0.4185**.

2 Introduction

We frame the problem from both a clinical relevance and a machine learning robustness perspective. Histological image classification plays a central role in computational pathology, supporting diagnostic decisions and treatment planning. The goal of this challenge is to classify microscopy images into four clinically relevant breast cancer subtypes.

The dataset presents several challenges: strong visual noise caused by pen markings and text anno-

tations, large background regions with no biological content, subtle visual differences between Luminal A and Luminal B, and a limited number of labeled samples. These factors significantly increase overfitting risk and encourage models to learn spurious correlations.

Instead of designing novel architectures, we focus on robustness through preprocessing, high-resolution training, and ensemble diversity. This strategy prioritizes generalization over model complexity and proved effective in this low-data regime.

3 Data Analysis and Preprocessing

Preprocessing decisions were guided by the goal of maximizing biologically meaningful signal while minimizing dataset-specific noise. Initial inspection revealed that a large portion of each image consists of irrelevant white background, which reduces effective resolution when resized.

3.1 Smart Cropping

We introduce a *Smart Cropping* procedure to maximize tissue coverage. Images are converted to grayscale, thresholded to identify non-white regions, cropped to the tissue bounding box, and padded to preserve aspect ratio. This removes approximately

30–50% of empty background, effectively increasing the resolution of informative tissue regions.

3.2 Input Resolution

All images are resized to 512×512 . Experiments with lower resolutions (e.g., 224×224) led to a significant performance drop (< 0.35), highlighting the importance of preserving fine-grained histological textures.

3.3 Augmentation

We apply heavy data augmentation using `torchvision`:

- Horizontal and vertical flips ($p = 0.5$)
- Rotations of 90° , 180° , 270°
- Color jitter on brightness, contrast, and saturation

These transformations are biologically valid, as histological images have no canonical orientation.

4 Model Architectures

All selected architectures are well-established backbones with complementary inductive biases. Our guiding principle is **ensemble diversity**. Different architectural biases produce partially uncorrelated errors, which improve ensemble robustness when averaged.

4.1 EfficientNet-B3

EfficientNet-B3 provides an effective balance between capacity and efficiency. It serves as our baseline architecture and achieves approximately 0.38 public score.

4.2 DenseNet-121 and DenseNet-169

DenseNets promote feature reuse and stable gradient flow. In practice, they act as strong regularizers and correct overconfident predictions from residual architectures.

4.3 ResNet-50 and ResNeXt-50

ResNet-50 offers reliable performance, while ResNeXt-50 (32x4d) introduces grouped convolutions to increase representational diversity. These models form the backbone of our heavier “Titan” experiments.

5 Training Strategy

Particular care was taken to ensure training stability and reproducibility across different random seeds. Training follows a *survival-of-the-fittest* paradigm, retaining only strong checkpoints across multiple random seeds.

5.1 Optimization

- Optimizer: AdamW with weight decay 1×10^{-3}
- Scheduler: OneCycleLR with peak learning rate 1×10^{-4}
- Loss: Cross-Entropy with label smoothing ($\epsilon = 0.1$)

Label smoothing reduced over-confident predictions on ambiguous samples, particularly between Luminal A and Luminal B. Training typically converged within 20–25 epochs.

5.2 Grand vs. Titan Ensembles

Two complementary streams are explored:

- **Grand Ensemble:** EfficientNet + DenseNet + ResNet (score 0.4118)
- **Titan Ensemble:** ResNeXt + DenseNet169 (score 0.4058)

Although lighter models perform better individually, merging both streams yields the strongest result, confirming the importance of error diversity.

6 Failed Experiments

Reporting unsuccessful strategies provides insight into the limitations of advanced techniques in this domain.

6.1 Mixup

Mixup degraded performance, likely because blending tissue textures produces non-physiological patterns that blur sharp cellular boundaries critical for diagnosis.

6.2 Pseudo-Labeling

Pseudo-labeling reduced the public score from 0.418 to 0.399 due to confirmation bias. Confident but incorrect predictions were reinforced during retraining, indicating that the baseline accuracy was insufficient for reliable semi-supervised learning.

7 Final Omni-Ensemble

The ensemble design reflects an incremental accumulation of improvements observed throughout experimentation. The final model aggregates 23 checkpoints:

- 4× EfficientNet-B3
- 8× DenseNet-121
- 4× ResNet-50
- 4× ResNeXt-50
- 3× DenseNet-169

Checkpoints were collected across different seeds and training recipes. During inference, all models contribute equally, as uniform averaging proved more stable than heuristic weighting.

7.1 Inference with TTA

We apply 8-view dihedral Test-Time Augmentation:

$$\{0^\circ, 90^\circ, 180^\circ, 270^\circ\} \times \{\text{original}, \text{flip}\}$$

Each image produces $23 \times 8 = 184$ probability vectors, averaged to obtain the final prediction. This yields a consistent performance gain of approximately +0.015.

8 Conclusion

Overall, the results highlight the effectiveness of systematic engineering choices in challenging low-data scenarios. A public leaderboard score of **0.4185** is achieved through robust preprocessing, ensemble diversity, and extensive test-time augmentation. The Omni-Ensemble demonstrates that combining many moderately strong models is more effective than relying on a single high-capacity model in low-data, high-variance settings.

8.1 Future Work

Future work includes stain normalization techniques and transformer-based architectures to capture long-range tissue structure. Cross-validation strategies may further improve reliability.